



The Lifeline – PSUs

The literal energy source for your 2025 PC build

A power supply unit (PSU) might not have RGB fans or benchmark charts, but it is the heart of any computer. It converts alternating-current (AC) power from your wall into the direct-current (DC) voltages needed by the motherboard, CPU, graphics card and peripherals. A reliable PSU not only feeds stable power but also protects components from voltage spikes, short circuits and over-current situations. In 2025, with powerful processors, multi-GPU setups and increasingly power-hungry AI accelerators, choosing the right PSU has never been more important.

Understanding the PSU's role

At its core, the PSU is a converter and regulator. It takes high-voltage AC electricity and transforms it into 3.3 V, 5 V and 12 V DC outputs. The unit must deliver clean, steady power to every component so that the system boots reliably and remains stable under load.

Modern PSUs include built-in protections such as short-circuit protection (SCP), over-voltage protection (OVP) and over-power protection (OPP) to guard against damage. High-quality units also provide energy efficiency and minimal heat generation, meaning less strain on cooling systems and lower electricity bills.

Built-in safety features

Reliable PSUs come equipped with multiple safety circuits. These include:

- **Short-circuit protection (SCP):** prevents catastrophic failures if a component shorts out, cutting power automatically.
- **Over-voltage protection (OVP):** triggers when output voltages exceed safe limits, protecting delicate components.
- **Over-power protection (OPP)/over-current protection (OCP):** shuts down the PSU if total power draw or current exceed rated limits.